

Measures of Central Tendencies

- **Measures of Central Tendency**: One number that represents a data set and gives an idea of the middle quality of the data.

(Measures of Central tendency show the tendency of some central value around which data tends to cluster)

There are three absolute measures of Central tendency

(1) Mean (2) Median (3) Mood

Mean: The mean can again be of three types

1. Arithmetic Mean: Arithmetic Mean is the sum of a set of observations (Positive, negative, zero) divided by the number of observations.

A.M.= Sum of values of items / Number of items

➤ **Direct Method of Calculation of A.M.:**

(I)Ungrouped data: Suppose we have a set of values $x_1, x_2, x_3, \dots, x_n$. Then we define and denote A.M. as

$$(a) \bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n} \text{ where } x_i \text{ is the individual values of items}$$

Example1: Find the mean of 10,20,30,50 and 60

Solution: A.M. = $\frac{10 + 20 + 30 + 50 + 60}{5} = \frac{175}{5} = 34$

(b) **Discrete Series**: When actual size of items (individual measurements) and corresponding frequencies are given.]]

$$\bar{x} = \frac{\sum_{i=1}^n (f_i x_i)}{\sum_{i=1}^n f_i = n}$$

Where x_i = the size of the item

and f_i = Corresponding frequency

Example 2: Calculate the arithmetic average:

Size of item(in inches)	2	3	6	7	10
Frequencies	3	8	5	10	14

Solution:

Size of item(in inches) (x_i)	2	3	6	7	10	
Frequencies (f_i)	3	8	5	10	14	$\sum f_i = 40$
$f_i x_i$	6	24	30	70	140	$\sum_{i=1}^5 (f_i x_i) = 270$

$$\text{A.M.} = \frac{270}{40} = 6.75 \text{ inches}$$

(I) Grouped data: i.e When class- interval and frequencies are given

$$\bar{x} = \frac{\sum_{i=1}^n (f_i x_i)}{\sum_{i=1}^n f_i = n}$$

Where x_i = the central value of the interval

and f_i = Frequencies of the respective classes

Example 3: Calculate the mean of the following distribution

Yield(in cwt)	0-3	3-6	6-9	9-12
Frequency	4	4	32	10

Solution:

Yield(in cwt)	0-3	3-6	6-9	9-12	
Central value(x_i)	1.5	4.5	7.5	10.5	
Frequency (f_i)	4	4	32	10	$\sum f_i = 50$
$f_i x_i$	6	18	240	105	$\sum_{i=1}^4 (f_i x_i) = 369$

$$\text{A.M.} = \frac{\sum_{i=1}^4 (f_i x_i)}{\sum_{i=1}^4 f_i} = \frac{369}{50} = 7.308 \text{ cwt}$$

H.W: 6: The following table gives the distribution of weekly wages of workers in a factory. Calculate the arithmetic mean of the distribution.

Weekly Wages:	240-269	270-299	300-329	330-359	360-389
No of Workers :	7	19	27	15	18

Advantages of Arithmetic Mean:

- (1) Easily Understand, calculated and is generally used
- (2) Uses each and every item

Disadvantages of Arithmetic Mean:

- (1) Its value may be greatly distorted by extreme value
- (2) Cannot be used in qualitative value

2.Geometric Mean: Ungrouped data:

The GM [G] of n-positive values $x_1, x_2, x_3, \dots, x_n$ is defined as the nth positive root of the product of n numbers

$$G = (x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n)^{1/n} = \left(\prod_{i=1}^n x_i \right)^{1/n} = \sqrt[n]{x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n}$$

$$\text{For frequency, } G = \sqrt[n]{x^{f_1}_1 \cdot x^{f_2}_2 \cdot x^{f_3}_3 \cdot \dots \cdot x^{f_n}_n}$$

where x_i is the individual values of items

and f_i = Corresponding frequency

Using Logarithm:

$$\text{Log } G = \frac{1}{n} (\log x_1 + \log x_2 + \dots + \log x_n)$$

$$= \frac{1}{n} \sum \log x$$

$$G = \text{Antilog} \left(\frac{1}{n} \sum \log x \right)$$

Or

$$\ln G = \frac{1}{n} \sum_{i=1}^n \ln x_i.$$

The latter expression states that the log of the geometric mean is the arithmetic mean of the logs of the numbers.

[Practical definition: The average of the logarithmic values of a data set, converted back to a base 10 number.]

Because the geometric mean is based on log values and the log transformation tends to draw extreme values toward the center of the data, the geometric mean is more "robust" than the arithmetic mean. "Robust" here means less influenced by outliers.

Grouped data: =

$$G = \sqrt[n]{X^{f_1}_1 X^{f_2}_2 X^{f_3}_3 \dots X^{f_n}_n} \quad \text{Where } x_i = \text{the central value of the interval}$$

$$f_i = \text{Frequencies of the respective classes}$$

$$\begin{aligned} \log G &= 1/N \log (X^{f_1}_1 X^{f_2}_2 X^{f_3}_3 \dots X^{f_n}_n) \\ &= 1/N [f_1 \log x_1 + f_2 \log x_2 + \dots + f_n \log x_n] \\ &= 1/N \sum f_i \log x_i \\ G &= \text{antilog} (1/N \sum f \log x) \end{aligned}$$

3. Harmonic Mean: $H.M = \frac{1}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$

For frequency,

$$\begin{aligned} H &= \frac{N}{\frac{f_1}{x_1} + \frac{f_2}{x_2} + \frac{f_3}{x_3}} \\ &= \frac{N}{\sum f/x} \end{aligned}$$

Where, x_i = individual values of item (For **Ungrouped data**)

x_i = the central value of the interval (For **Grouped data**)

and f_i = Frequencies of the respective classes

- Home

Property: For set of n non-negative observations $x_1, x_2, x_3, \dots, x_n$

$$AM \geq GM \geq HM$$

Let 3, 9, 11 are three given numbers, then

$$AM = (3+9+11) / 3 = 7.66$$

$$GM = \sqrt[3]{3 \cdot 9 \cdot 11} = 6.67$$

$$HM = \frac{3}{\frac{1}{3} + \frac{1}{9} + \dots + \frac{1}{11}} = 5.60$$

So $AM \geq GM \geq HM$ is proved

- **Median:** The “Median” is the middle value in the list of numbers.

Calculation of median: Ungrouped Data

Arrange the data in ascending or descending order of magnitude.

$$\text{Median} = \text{Size of } \frac{n+1}{2} \text{ th observation}$$

For Uneven ,

Ex: from the following data of wages of 7 workers compute the median wage

Wages (in Taka) 1600 1650 1580 1690 1660 1606 1640

Solution: wage arranged in ascending order

1580, 1600, 1606, 1640, 1650, 1660, 1690

$$\text{Median} = \text{Size of } \frac{n+1}{2} \text{ th observation} = \frac{7+1}{2} \text{ th observation} = 4 \text{ th observation}$$

Here median wage = Taka 1640.

For Even,

$$\frac{8+1}{2} \text{ th observation} = 4.5 \text{ th observation}$$

We shall take 4th and 5th observation

$$\text{Median} = \frac{1640 + 1650}{2} = 1645$$

Advantages :(1) Its value is not distorted by extreme values.

(2) It is suitable for qualitative studies.

Disadvantages: (1) Necessities arraying of data before it can be found.

Grouped data:

$$\text{Median} = L + \frac{\frac{N}{2} - p.c.f}{f} i$$

Where,

L= Lower limit of median class

P.c.f= preceding cumulative frequency to the median class

f= Frequency of median class

i= The class interval of median class

$$\text{Median class, CF(cumulative frequency)} \geq \frac{N}{2}$$

Ex. 1500 workers are working in an industrial establishment. Their age is classified as follows:

Age(yrs)	No.of workers	Age (yrs)	No.of workers
18-22	120	38-42	184
22-26	125	42-46	162
26-30	280	46-50	86
30-34	260	50-54	75
34-38	155	54-58	53

Calculate the median age

Solution:

Age group	Frequency (f_i)	Cumulative frequency(c.f)
18-22	120	120
22-26	125	245
26-30	280	525
30-34	260	785
34-38	155	940
38-42	184	1124
42-46	162	1286
46-50	86	1372
50-54	75	1447
54-58	53	1500

So, Median = Size of $\frac{N}{2}$ th observation = Size of $\frac{1500}{2}$ th observation = 750 th observation

Hence of median lies in the class 30-34

$$\text{Median} = L + \frac{\frac{N}{2} - p.c.f}{f} \cdot i = 30 + \frac{750 - 525}{260} \times 4 = 30 + 3.46 = 33.46 \text{ (Answer)}$$

- **Mode:** Mode is the most frequently occurring value in a frequency distribution.

Ex. To find the mode of 11,3,5,11,7,3,11

Here, mode=11

Calculation of mode –Ungrouped data

For example, let us consider the age of (in year) of some children investigated in a small locality:

5, 2, 2, 8, 7, 6, 5, 4, 3, 4, 5, 2, 2

In the above example, age 2 years are recorded 4 times (maximum time). So 2 years is the mode of the distribution of ages of children.

Distribution of workers by their daily wages (in taka):

90.00, 90.00, 95.00, 100.00, 100.00, 90.00, 100.00, 110.00, 115.00, 125.00

Here 90.00 and 100.00 occur with greatest frequency 3.

Therefore, Mo=90.00 and Mo=100.00. The distribution is bi-modal.

Distribution of the students according to their grade point average in an examination:

2.5, 3.0, 3.0, 3.5, 3.8, 3.0, 3.0, 2.5, 3.8, 4.0, 4.0, 4.0, 4.0, 2.5, 2.5

In the distribution $M_o=2.5$, $M_o = 3.0$ and $M_o= 4.0$, since each of the value 2.5, 3.0 and 4.0 occur 4 times in the distribution. This distribution is known as multimodal distribution.

Calculation of mode –Grouped data

Example: the following data represent the distribution of female workers in different garments industries garments industries according to their monthly salary (in taka).

Class interval of salary (in taka)	No. of female workers f_i	c.f
< 600	10	10
600-700	40	50
700-800	65	115
800-900	250	365
900-1000	175	540
1000-1100	82	622
1100-1200	50	672

- Find the maximum salary, on average, of the major group of female workers.
- How many female workers have salary less than 1000 taka?

Solution: (i) the major group of of female workers are 250 whose salary is in the limit 800-900. Their, on an average, salary is given by mode, where

$$\begin{aligned}
 M_o &= l + \frac{h(f_1 + f_0)}{2f_1 - f_0 - f_2} \\
 &= 800 + \frac{100(250 - 65)}{2 \times 250 - 65 - 175} \\
 &= 871.15 \text{ tk}
 \end{aligned}$$

$$l = 800, h = 100, f_1 = 250, f_0 = 65, f_2 = 175$$

(ii) From c.f it is observed that 540 female workers' salary is less than 1000.00 tk.

An empirical relationship between mean, median, mode is presented by Karl Pearson. For moderately asymmetrical curve, the empirical relationship is

$$\text{Mode} = \text{Mean} - 3(\text{Mean} - \text{Median})$$

$$\text{Or Mode} = 3\text{Median} - 2\text{Mean}$$

$$\text{Median} = \text{Mode} + \frac{2}{3} (\text{Mean} - \text{Mode})$$

$$\text{And Mean} = \text{Mode} + \frac{3}{2} (\text{Median} - \text{Mode})$$

Example: The following data represent the distribution of chicken by weight after 7 weeks of birth:

Class interval of weight(in kg)	No. of chicken, f_i	c.f	Mid-Value X_i	$f_i X_i$
0.6-0.8	25	25	0.7	17.5
0.8-1.0	22	47	0.9	19.8
1.0-1.2	18	65	1.1	19.8
1.2-1.4	12	77	1.3	15.6
1.4-1.6	12	89	1.5	18.0
1.6-1.8	11	100	1.7	18.7
Total	100			109.4

Find the Weight, on average, of the maximum number of the chicken.

Solution: Since the maximum frequency is in the first class, mode is ill defined.

However, $\text{Mean} = \frac{1}{N} \sum f_i X_i = \frac{109.4}{100} = 1.094$

$$\text{Me} = l + \frac{h}{f} \left(\frac{N}{2} - c \right) = 1.0 + \frac{0.2}{18} (50 - 47) = 1.03$$

The weights, on average, of maximum number of chicken is mode of the distribution.

The mode is given by, $\text{Mo} = 3 \text{ Median} - 2 \text{ mean} = 3(1.03) - 2(1.094) = 0.902$

Outlier:

- A number that is very far away from the rest of the data set. I.e., a value that is much greater or much less than most of the other numbers
- An outlier moves the average from the middle of the cluster of the rest of the data points.

Advantage & Disadvantage of Measure Of Central Tendency

- Consider a set of observations as follows :
- Observations: 4, 8, 5, 7, 100

$$\bar{x} = \frac{4+8+7+100}{5} = 24.8$$

Disadvantage of Arithmetic Mean :

Arithmetic mean is affected by extreme value. This is the major disadvantage.

Array: 4, 5, 7, 8, 100

Me = 7

Advantage of Median:

Me is not affected by extreme value

X : -2, 4, 8

$$G = (-2.4.8)^{1/3}$$

If we have even number of minus values i.e. X: -2, 4, -8

$$G = (-2.4.-8)^{1/3} = 4 \text{ and } \bar{x} = -2$$

But here $G > \bar{x}$, it's distorted the relationship, So for negative values we shouldn't calculate Geometric mean

X: 3, 4, 0, 10

$$G = (3 \cdot 4 \cdot 0 \cdot 10)^{1/4}$$

$$= 0$$

$$H = \frac{4}{1/3 + 1/4 + 1/0 + 1/10}$$

$$= \frac{4}{\infty} = 0$$

Disadvantage of Geometric & Harmonic Mean:

Geometric & Harmonic mean are not calculated if array observation is Zero or negative.

- Geometric & Harmonic mean are used if observations are given in rates & ratio.

X: 2, 3, 3, 5, 8, 9, 15, 15

Mode = 3 & 15

Class interval	Frequency	Frequency
10-20	$f_0 = ?$ $20 = f_1$	8
20-30	$18 = f_2$	10
30-40	14	12
40-50	10	$14 = f_0$
50-60	6	$18 = f_1$

$$f_2 = ?$$

*** Modal Class is that class for which frequency is maximum.

Disadvantage of Mode :

1. There may be many modes
2. Mode may not fall at the centre of the array.
3. Mode is not well defined if the modal class is the first class & last class.

Example:-

Find an appropriate measure of central tendency of the following observations with justification.

Ans :-

1. Here arithmetic mean is not suitable as there is an extreme value 150.
2. Geometric & Harmonic mean are not calculated as there are zero & negative observations.
3. Each observation occurs once. So there is no mode.
4. Therefore median is an appropriate measure of central tendency in this case.

$$Me = \frac{0+5}{2} = 2.5$$



$$Me = \frac{-10+10}{2} = -5$$

Measures of Location

Let us, consider the array as follows

X: 5, 7, 8, 10, 13, 15, 16, 18, 19, 20, 25, 27, 29, 31, 33, 36, 37, 38, 40, 45

N = 20

$$\text{Me} = \frac{13+15}{2}$$

$$= 14$$

$$\text{Me} = \frac{20+25}{2}$$

$$= 22.5$$

$$\text{Me} = \frac{33+36}{2}$$

$$= 34.5$$

A measure which is located in different place in the array is called the measure of location.

- Measure of **central tendency** falls at the **centre of the array** but measure of **location** falls in **different places** in the array.

Different measures of location are :

1. Quartile (Q_i $i= 1, 2, 3$)
(A measure which divides the array into 4 equal parts)
2. Decile (D_i , $I = 1, 2, \dots, 9$)
(A measure which divides the array into 10 equal parts)
3. Percentile (P_i ; $I = 1, 2, \dots, 99$)
(A measure which divides the array into 100 equal parts)

$$\begin{aligned}\text{Me} &= l + \frac{h}{f} \left(\frac{N}{2} - c \right) = Q_2 = D_5 = P_{50} \\ &= l + \frac{h}{g} \left(\frac{2N}{4} - C \right) \\ &= l + \frac{h}{g} \left(\frac{5N}{10} - C \right) \\ &= l + \frac{h}{g} \left(\frac{2N}{4} - C \right) \\ &= l + \frac{h}{g} \left(\frac{50N}{100} - C \right)\end{aligned}$$

$$Q_i = l + \frac{h}{f} \left(\frac{iN}{4} - C \right); i = 1, 2, 3$$

Here,

l = lower limit of Q_i class

h = width of Q_i class

f = frequency of Q_i class

c = c.f of the class before Q_i class

N = Total frequency

* Q_i is that class for which $c.f. \geq \frac{iN}{4}$

$$### D_i = l + \frac{h}{f} \left(\frac{iN}{10} - c \right); i = 1, 2, \dots, 9$$

* D_i is that class for which $c.f. \geq \frac{iN}{10}$

$$P_i = l + \frac{h}{f} \left(\frac{iN}{100} - c \right), i = 1, 2, \dots, 99$$

* P_i is that class for which $c.f. \geq \frac{iN}{100}$

Let us consider the following frequency distribution:

Class interval	Frequency ,f	C.f.
10-12	8	8

12-14	12	20
14-16	20	40
16-18	22	62
18-20	18	80
20-22	12	92
22-24	8	100
	N = 100	

1. Calculate Q_1 , Q_2 , Q_3 Of the distribution
2. Find the values of D_3 & D_7
3. Find the values of p_{40} & p_{80}
4. Find Q_1 , D_8 & P_{60} from graph
5. Draw Box & whisker plot of the distribution

$$1. \quad Q_1 = l + \frac{h}{f} \left(\frac{N}{4} - c \right)$$

$$= 14 + \frac{2}{20} \left(\frac{100}{4} - 20 \right)$$

$$= 14 + \frac{1}{10} \times 5$$

$$= 14.5$$

$$Q_2 = l + \frac{h}{f} \left(\frac{2N}{4} - c \right)$$

$$= 16 + \frac{2}{22} \left(\frac{2 \times 100}{4} - 40 \right)$$

$$= 16 + \frac{1}{11} \times 10$$

$$= 16.9$$

$$Q_3 = l + \frac{h}{f} \left(\frac{3N}{4} - c \right)$$

$$= 18 + \frac{2}{18} \left(\frac{3 \times 100}{4} - 62 \right)$$

$$= 18 + \frac{1}{9} \times 13$$

$$= 19.44$$

2.

$$D_3 = l + \frac{h}{f} \left(\frac{3N}{10} - c \right)$$

$$= 14 + \frac{2}{20} \left(\frac{3 \times 100}{10} - 20 \right)$$

$$= 14 + \frac{1}{10} \times 10$$

$$= 15$$

$$D_7 = l + \frac{h}{f} \left(\frac{7N}{10} - c \right)$$

$$= 18 + \frac{2}{18} \left(\frac{7 \times 100}{10} - 62 \right)$$

$$= 18 + \frac{1}{9} \times 8$$

$$= 18.9$$

Box Plot

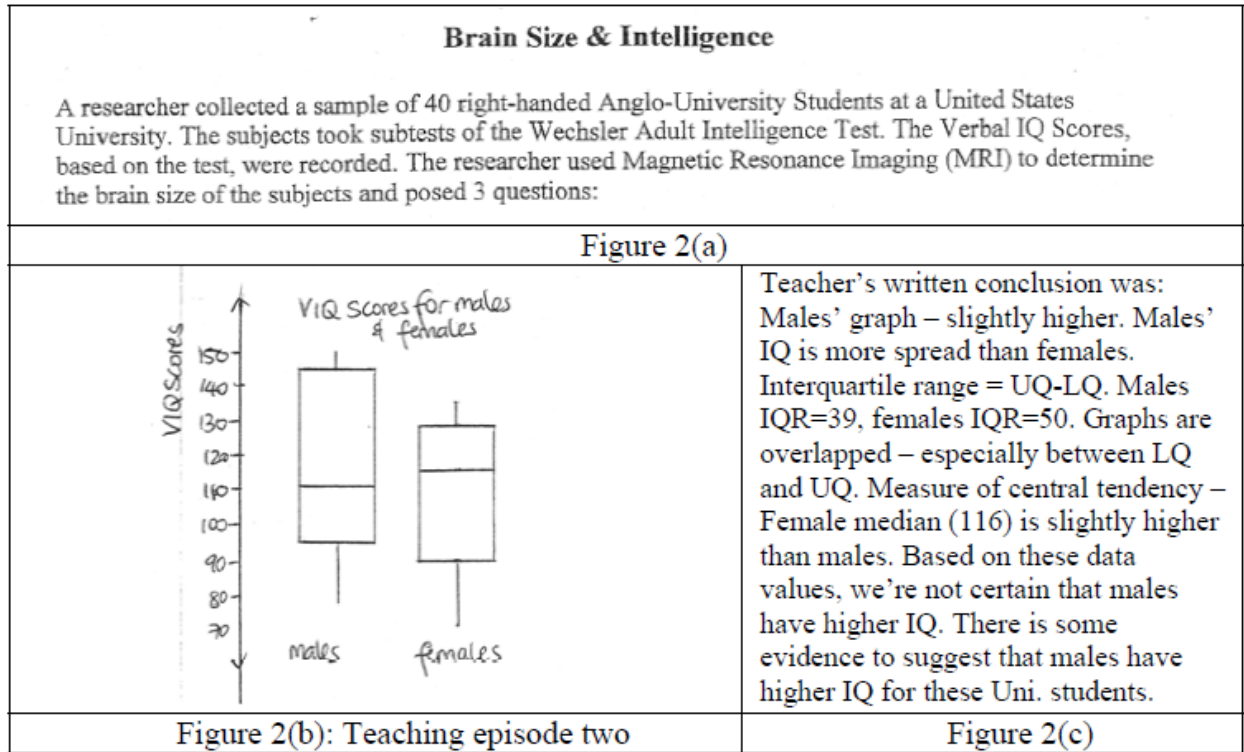


Figure 2. Teaching episode two – graph given to students by teacher

Outlier: An outlier is defined as a value that is more than 1.5 times the interquartile range smaller than Q1 or larger than Q3.

$$\text{Outlier} > Q3 + 1.5(Q3 - Q1)$$

$$\text{Outlier} < Q1 - 1.5(Q3 - Q1)$$

Dispersion

The word 'Dispersion' implies the average distance or deviation from a central value.

Let us start with two examples

Data set-1: 3,4,5 mean=4

Data set-2: 0,2,10 mean=4

Which implies same mean but in set 2 much variation

The absolute measures are

- 1) The Range (2) The Standard deviation (3) The Mean Deviation (4) The quartile deviation

The relative measures are

- 1) The coefficient of variation
- 2) The coefficient of mean deviation
- 3) The coefficient of Quartile deviation
- 4) The coefficient of Range

*** **Absolute measure's** of variation are expressed in the same statistical unit in which the original data are given such as rupees, kilograms, tones etc.
i.e. when two distributions have same unit.

*** **Relative measures of variation:** A measure of relative variation is the ratio of a measure of absolute variation to an average. It is sometimes called a coefficient of variation, because “Coefficient” means a pure number that is independent of the unit of measurement.

Range: Range is the difference between the value of smallest and largest.

Symbolically, $\text{Range} = L - S$ where, L = Largest value and S = Smallest

- The range is widely used in statistical process control applications.
A serious defect of the range is that it is based on only two values, the largest and smallest; it does not take into consideration all of the values

Limitations:

- 1) Range is not based on each and every observation.
- 2) The amount of range is affected by extreme values
- 3) It is not calculated from frequency distribution with open-end classes.

Mean deviation: Mean deviation is an average of absolute deviation of the individual observation from the central value of the series.

$$\text{MD} = \frac{|x_1 - \bar{x}| + |x_2 - \bar{x}| + \dots + |x_n - \bar{x}|}{n} = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n}$$

Ex. Find the mean deviation of 4, 8, 12, 24

Average $\bar{x} = (4+8+12+24)/4 = 12$

$$\text{MD} = \frac{|4-12| + |8-12| + |12-12| + |24-12|}{4} = 6$$

Limitations:

- 1) It ignores the sign in measuring deviations which is bad from the mathematical point of view. Hence, it is not good for further algebraic use of the measure.
- 2) The amount of mean deviation increases with the increase in size of sample.

Standard Deviation: (<http://www.mathsisfun.com/data/standard-deviation.html>)

It is a measure of how much “Spread” or “variability” is present in the sample.

[NB. If all the numbers in the sample are very close to each other, the standard deviation is close to zero.]

Standard deviation is denoted and is defined as

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}} \quad \leftarrow \text{For population}$$

Where, x_i = the individual values

μ = Population mean

n = number of items

$\therefore$ Variance σ^2 = (Standard deviation)²

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}} \quad \leftarrow \text{For Sample}$$

For Frequencies:

$$\sigma = \sqrt{\frac{\sum_{i=1}^n f_i (x_i - \mu)^2}{n}} \quad \leftarrow \text{For population}$$

Where, x_i = mid value of each class interval (For grouped data)

$$s = \sqrt{\frac{\sum_{i=1}^n f_i (x_i - \bar{x})^2}{n-1}} \quad \leftarrow \text{For Sample}$$

Advantage:

- 1) Based on all the observations.
- 2) As sign of deviations are not ignored, it is suitable for further use in statistical analysis.

Limitations:

As it is not free of unit, it is not use to compare the dispersion of two or more distributions.

Ex. Find the standard deviation from a population data on the weekly wages of ten workers working in a factory.

Wages(TK): 320,310,315,322,326,340,325,321,320,331

$$\mu = \frac{\sum_{i=1}^{10} x_i}{10} = \frac{3230}{10} = 323 \quad \therefore \sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}} = \sqrt{\frac{622}{10}} = 7.89$$

Quartile deviation:

The inter quartile range is frequently reduced to the measure of semi-inter-quartile range, also known as the quartile deviation, by dividing it by 2. Thus

$$Q_D = \frac{I_{QR}}{2} = \frac{Q_3 - Q_1}{2}$$

This measure is more meaningful than the range because it is not based on two extreme values.

Limitation of Quartile Deviation as a Measure of Dispersion:

the quartile deviation have serious shortcomings. First of all, they do not take into consideration the values of all times. For example, Q_D is not affected by the distribution of all items below Q_1 and above Q_3 . Moreover, they remain to be positional measures, following to provide measurement of scatter of the observations, relative to the typical value. In addition, it does not enter into any of the higher mathematical relationships that are basic to inferential statistics.

Relative measures of dispersion:

Coefficient of range:

The coefficient of range is a relative measure corresponding to range and obtained by the following formula

$$C_R = \frac{L - S}{L + S} 100$$

Where L and S are respectively the largest and smallest observations in the data set. The coefficient of range is rarely used as a measure of dispersion because of its inherent difficulties in interpretation.

Coefficient of Mean Deviation:

The third relative measure is the coefficient of mean deviation. As the mean deviation can be computed from mean, median, mode or from any arbitrary value, a general formula for computing coefficient of mean deviation may be put as follow:

$$C_{MD(a)} = \frac{M_d(a)}{a} 100$$

Where 'a' may be the mean, median, mode or any arbitrary value.

Coefficient of Quartile Deviation:

The coefficient of quartile deviation is computed from the first and third quartiles using the following formula:

$$C_{QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1} 100$$

It is worth to mention that most of the absolute measures except CV, are of little significance because of their limited practical utility.

Co-efficient of variation (CV):

A co-efficient of variation is computed as the ratio of the Standard deviation of the distribution to the mean of the same distribution.

$$CV = \frac{S}{\bar{x}} \times 100$$

Uses of co-efficient of variation: CV is helpful in comparing the relative variation in several data sets that have different means and different standard deviation.

EX.: Find CV for data given below:

For 150 children

Height	Weight
Mean, $\bar{x}$ = 40 inch	Mean, $\bar{x}$ = 10 kg
SD, S = 5 inch	SD, S = 2 kg

$$\text{For Height CV} = \frac{S}{\bar{x}} = \frac{5}{40} = 0.125$$

$$\text{For Weight CV} = \frac{S}{\bar{x}} = \frac{2}{10} = 0.2$$

So we can say weight variability is greater than height variability.

Ex. For Blood pressure:

Systolic	Diastolic
Mean, $\bar{x}$ = 130 mm Hg	Mean, $\bar{x}$ = 60 mm Hg
SD, S = 15 mm Hg	SD, S = 8 mm Hg

$$\text{For Systolic, CV} = \frac{S}{\bar{x}} = \frac{15}{130} = 0.115$$

$$\text{For Diastolic, CV} = \frac{S}{\bar{x}} = \frac{8}{60} = 0.133$$

Example: Suppose that we wish to obtain some insight into whether height is more variable than the weight in the same population. For this purpose, we have to following data obtained from 150 children in a community.

	Height	Weight
Mean	40 inch	10 kg
SD	5 inch	2 kg
CV	12.5%	20.0%

Examination of the respective standard deviations does not tell us in any meaningful way which characteristic has more variability than the other, because they are measured in different units. If we now compute coefficient of variation, the results become comparable, because coefficient of variation for weight is greater than that of the height, we conclude that weight has more variability than height in the population.

Even if two variables in the same population are measured in the same unit, the standard deviation may fail to provide a correct picture of their relative variability. This is illustrated by an example bellow

Example: Consider that the blood pressures of a group of patients were measured at tow level: systolic and diastolic, both being measured in the same unit. The results were as follow:

	Systolic	Diastolic
Mean	130 mm Hg	60 mm Hg
SD	15 mm Hg	8 mm Hg
CV	11.5%	13.3%

As implied by the standard deviation, systolic pressure is more than the diastolic pressure. However in relative terms, as measured by the CV, the diastolic pressure has the greater variability. This show the relative variability is of more concern than absolute variation hence the importance of the coefficient of variation.

The discussions and examples above tend to demonstrate that coefficient of variation is a very useful measure when:

- 1) **The data are in different units.**
- 2) **The data are in same units but means are far apart.**

3) When the data sets involve all or nearly all positive values.

Properties of the Standard Deviation

In terms of measuring the variability of spread of data, we've seen that the [standard deviation](#) is the preferred and most used measure.

Some additional things to think about the standard deviation:

1. The standard deviation is the typical or average distance a value is to the mean
2. If all values are the same, then the standard deviation is 0
3. The standard deviation is heavily influenced by outliers just like the mean (it uses the mean in its calculation).
4. The sample standard deviation is denoted with the letter s and the population standard deviation is denoted with the lower case Greek letter sigma σ .

If your data is more spread out (has more variability) then you will have a higher standard deviation. It's often difficult to interpret a standard deviation since it's based on the sample of data. Is a standard deviation of 12 high or is a .20 high?

Coefficient of Variation (CV)

If you know nothing about the data other than the mean, one way to interpret the relative magnitude of the standard deviation is to divide it by the mean. This is called the coefficient of variation. For example, if the mean is 80 and standard deviation is 12, the $cv = 12/80 = .15$ or 15%.

If the standard deviation is .20 and the mean is .50, then the $cv = .20/.50 = .4$ or 40%. So knowing nothing else about the data, the CV helps us see that even a lower standard deviation doesn't mean less variable data.